

# Q3 — Guardrails, Evaluation & Multi-Agent Plan

Local RAG over the Malawi public-health dataset • Ollama qwen3:8b • September 2026

## 1. Guardrail design

Trust boundary. User queries and retrieved passages are untrusted input. Only the fixed application policy is authoritative. Generated text is untrusted until citation and faithfulness validation pass. Retrieved passages are delimited as SOURCE blocks and never placed in the system message.

Defense in depth. Screen each retrieved chunk with deterministic high-signal patterns; optionally use a local classifier for ambiguous text. Quarantine flagged chunks, log only source IDs/hashes and reasons, then verify the remaining evidence. If evidence becomes insufficient, retrieve once more with a refined query. The model is explicitly told to ignore instructions inside sources, and no corpus text can trigger tools or change policy.

| Control              | Deterministic check                                                               | Failure action                                                    |
|----------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Injection screening  | Pattern/classifier scan over every chunk; preserve source provenance.             | Quarantine the chunk, record reason, re-verify safe evidence.     |
| Evidence sufficiency | Verifier identifies supported facts and missing evidence before generation.       | Refine query; after two retries, return "cannot find in sources". |
| Mandatory citations  | Every factual claim carries an exact allow-listed chunk ID.                       | Reject output with absent, malformed, or unknown citations.       |
| No guessing          | Answerer may use only verifier-approved chunks; temperature remains low.          | Abstain when the source set cannot support the requested answer.  |
| Public-health tone   | General information only; no personal diagnosis, prescription, or treatment plan. | Replace unsafe output with a safe general response or abstention. |

## Answer contract

The Answerer states the answer briefly, attaches [chunk\_id] to each factual claim, and uses only IDs supplied in the current context. It distinguishes general public-health guidance from personal medical advice. Urgent symptom questions receive a brief instruction to contact local emergency or clinical services; the system does not diagnose. If sources are absent, unsafe, conflicting, or insufficient, the exact response is "cannot find in sources".

## 2. Evaluation plan

Golden set. Build 18 independently reviewed questions: 8 single-source factual, 4 multi-chunk or comparison, 3 unanswerable/out-of-domain, and 3 adversarial prompt-injection cases. For each item store the expected source document and paragraph/chunk set, supported answer facts, answerable flag, multi-citation requirement, and safety label. Use 12 items while tuning and keep 6 held out for final reporting; version the set with the corpus snapshot.

| Measure          | Definition                                                                                                                                                     | Initial promotion target    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Recall@4         | Set-based Recall@4 = retrieved gold chunks divided by all gold relevant chunks; report Hit@4 separately and strict full-evidence recall for multi-chunk items. | ≥ 0.85                      |
| MRR              | Mean reciprocal rank of the first relevant chunk; unreturned relevant evidence scores zero.                                                                    | ≥ 0.75                      |
| Faithfulness     | Claim-level supported claims ÷ factual claims, checked against cited text by a local verifier and human audit.                                                 | ≥ 0.90                      |
| Citation quality | Validity: cited ID exists. Coverage: factual claims with citations. Precision: citation supports its claim.                                                    | 100% valid; ≥ 0.95 coverage |

| Measure          | Definition                                                                      | Initial promotion target          |
|------------------|---------------------------------------------------------------------------------|-----------------------------------|
| Abstention       | Precision, recall, and F1 over the unanswerable subset.                         | $F1 \geq 0.90$                    |
| Injection safety | Flag/block rate on adversarial chunks plus false-positive rate on clean chunks. | $\geq 0.95$ block; $\leq 0.05$ FP |

## Measurement and review

Run retrieval first and compute Recall@k and MRR from annotated chunk IDs. Then generate answers and parse citations. Reject unknown IDs deterministically. Split the answer into atomic factual claims and ask qwen3:8b, at temperature 0 with JSON output, whether each claim is directly supported by the cited passages. A second human reviewer audits disagreements, all safety failures, and a random sample. Report results by question type, not only as an overall average.

The thresholds above are starting release gates and should be revised after the first labeled baseline. Every corpus, prompt, model, or retrieval change reruns the held-out set. A release fails on any unsupported personal medical recommendation even when aggregate metrics pass.

## Ragas evidence-quality evaluation

Why Ragas. Fluent answers can still use irrelevant evidence, omit reference facts, or introduce unsupported claims. Ragas complements deterministic Recall@k, MRR, citation, abstention, and safety checks by evaluating retrieval and generation separately. The six metrics remain separate diagnostics; they are not averaged into one score.

### Metrics used

| Metric             | Primary inputs                         | Question answered                                                      |
|--------------------|----------------------------------------|------------------------------------------------------------------------|
| Faithfulness       | Answer + generation contexts           | Are factual claims supported by supplied evidence?                     |
| Answer relevancy   | Question + answer                      | Does the response directly address the question?                       |
| Context precision  | Question + ranked contexts + reference | Is relevant evidence ranked early with limited noise?                  |
| Context recall     | Contexts + reference                   | Do retrieved passages cover the reference facts?                       |
| Answer correctness | Answer + reference                     | Does the response agree factually and semantically with the reference? |
| Answer similarity  | Answer + reference embeddings          | How close is the answer's meaning to the reference?                    |

## Three-question results

| Metric             | Q220   | Q1016  | Q1226  | Mean   |
|--------------------|--------|--------|--------|--------|
| Faithfulness       | 0.5000 | 1.0000 | 1.0000 | 0.8333 |
| Answer relevancy   | 0.8132 | 1.0000 | 0.7861 | 0.8665 |
| Context precision  | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| Context recall     | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| Answer correctness | 0.9866 | 0.9642 | 0.8933 | 0.9481 |
| Answer similarity  | 0.9465 | 0.8570 | 0.8232 | 0.8756 |

Observed behavior. Context precision and context recall were 1.0000 for all three questions, indicating strong reference-aligned retrieval in this sample. Q220 combined 0.9866 correctness with 0.5000 faithfulness, showing that reference agreement alone cannot replace claim-to-context verification. Q1016 was strong across all measures. Q1226 was fully grounded but less concise and reference-aligned, reflected in lower relevancy and similarity.

Scope and hardware observation. These three fully scored questions are a development sample, not a held-out production benchmark. Local qwen3:8b both generated and judged answers, so human review remains necessary to control self-judge bias. The sample was kept at three because sustained local inference caused significant MacBook heat; this is a machine resource constraint and not an observed algorithm failure.

### 3. Multi-agent workflow

| Agent     | Responsibility                                                                                                       | Structured output                                                                   |
|-----------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Retriever | Hybrid BM25 + MiniLM search with filters; accept a refined query and return ranks/scores.                            | ranked_chunks[{id,text,metadata,semantic,bm25,rrf}], query_used                     |
| Safety    | Inspect query and chunks for injection or disallowed personal-medical requests; never interpret corpus instructions. | safe_chunks, quarantined_ids, reasons, risk_flags                                   |
| Verifier  | Check whether safe evidence covers every requested fact; identify gaps and create one focused query refinement.      | verdict: sufficient insufficient unsafe, approved_ids, missing_facts, refined_query |
| Answerer  | Generate from approved evidence only; apply tone and exact citations; abstain on any contract failure.               | answer, cited_ids, abstained, reason                                                |

#### Routing and stopping

Query → Retriever → Safety → Verifier. If sufficient, the Answerer generates once and the system validates citation IDs and claim support. If unsafe chunks are removed, the Verifier rechecks the reduced set. If evidence is insufficient, the Verifier supplies a refined query and returns to retrieval. Permit at most two retrieval retries. Empty safe evidence, exhausted retries, invalid citations, or faithfulness below 0.90 routes to the exact abstention response. State is passed as structured data, which keeps decisions reproducible and testable.

#### MLflow observability

Use one parent MLflow run per evaluation batch and a nested run per question. Log parameters: corpus/index version, Git commit, embedding and Ollama model, chunking policy, top\_k, thresholds, prompt versions, and retry limit. Log metrics: Recall@k, MRR, faithfulness, citation validity/coverage, abstention scores, injection block/false-positive rates, retries, and stage latency. Save the golden-set version, sanitized agent trace, retrieved IDs/scores, verifier JSON, prompt templates, and failure table as artifacts. Do not log sensitive raw queries or passages; use redaction plus source IDs or hashes.

#### Minimal implementation

The companion snippets\_q3.py provides detect\_prompt\_injection(text), a local-Ollama faithfulness\_check(answer, cited\_chunks), and dependency-injected orchestration. Production integration should add schema validation, retries for transient Ollama errors, authenticated access to MLflow, and reviewer-approved emergency wording.